



[Integrated Energy Systems Office](#) > | [Agrivoltaics: Solar and Agriculture Co-Location](#)

# Agrivoltaics: Solar and Agriculture Co-Location

## What is Agrivoltaics?

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Most large, ground-mounted solar photovoltaic (PV) systems are installed on land used only for solar energy production. However, it is possible to co-locate solar systems and agriculture on the same land. This practice, also known as agrivoltaics or dual-use solar, involves locating agricultural production, such as crops, livestock, or pollinator habitats, underneath solar panels or between rows of solar panels.

Exploring methods that optimize both energy and agricultural production at co-located sites is an active area of research both at DOE and other government agencies like the U.S. Department of Agriculture. This research may offer opportunities to increase overall land productivity. Research efforts aim to quantify the potential effects on farmer revenue, ecosystems, rural communities, and [solar energy soft costs](#)—the non-hardware costs of solar energy deployment. Understanding these dynamics can help to evaluate the benefits and challenges of this growing practice.

# Why is Agrivoltaics Research Important?

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Agrivoltaics research provides information to enable farmers, ranchers, and other agricultural producers to determine whether they could gain value from installing solar systems while keeping their land available for agricultural purposes. Since installing a solar energy system is a long-term investment with an expected system lifetime of 25 years or more, having accurate research and analysis available will help these stakeholders make informed decisions.

Agrivoltaics research has shown that the co-location of solar PV and agriculture **could provide agricultural enterprises with benefits such as diversified revenue sources and ecological advantages, while reducing land use competition** and siting restrictions. Research in agrivoltaics explores how proper planning, mitigation, and construction techniques can reduce environmental impacts of large-scale solar installations. This research also investigates how to keep land under and around the system available for agricultural production, including pollinator habitats, livestock grazing, and crop production.

## SETO Research in Solar and Agriculture Co-Location

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SETO projects in this research topic evaluate practices, develop technologies, and conduct research and analysis to maximize the value that co-located solar technologies could provide while keeping land available for agricultural purposes.

The [Foundational Agrivoltaic Research for Megawatt Scale \(FARMS\) funding program](#) funds projects that optimize agrivoltaics designs, integrate agrivoltaics into existing solar systems, and help lower the barriers of entry to agrivoltaics. Additionally, the [Solar Energy Technologies Office Fiscal Year 2020 funding program](#) funds projects that develop new system designs and technologies, build profitable co-location models, and examine the impacts of solar and agriculture co-location.

## Additional Resources

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- [Pollinator Habitat Aligned with Solar Energy \(PHASE\)](#).<sup>☞</sup>
- [Summary: Solar Impacts on Wildlife and Ecosystems Request for Information](#)
- [AgriSolar Clearinghouse](#) <sup>☞</sup>: an information-sharing, relationship-building, public communications hub for agrivoltaics
- [Buzzing Around Solar: Pollinator Habitat Under Solar Arrays](#)

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